

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
2.	(a)	(i)		5 points plotted correctly < 1 small square (2) 4 points plotted correctly < 1 small square (1) 3 or less points plotted correctly < 1 small square (0) Straight line between 1.0 – 5.0 cm wavelength < 1 small square (1)		3		3	3	3
		(ii)	I	Increases		1		1		1
			II	Decreases		1		1		1
		(iii)	I	2[.0] [cm]		1		1		1
			II	$v = 2.0 \text{ ecf} \times 10$ (1) $v = 20$ [cm/s] (1)	1	1		2	2	2
	(b)	(i)		Transverse (1) Geostationary (1) Microwaves (1) Equator (1)	4			4		
		(ii)	I	Distance = 264 789.9 [km] OR 264 924.2 [km]		1		1	1	
			II	Substitution: $\frac{264\,924.2 \text{ ecf}}{24}$ (1) Speed = 11 032.9 [km/h] OR 11 038.5 [km/h] (1) Award 1 mark for an answer of 1 756.8 [km/h] using orbit radius if no answer in (ii)I	1	1		2	2	

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		(iii)		Satellite has further to travel (1) so must travel faster / in the same time / in 24 hours (1) [so she is incorrect]			2	2		
				Question 2 total	6	9	2	17	8	8